Improve and Challenge

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Improve and Challenge

> Prepare Improve and Challenge: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Repair a specific observed failure

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Test Diverse Users

Without opening the old lesson, explain one key idea from Test Diverse Users. Give an example and a mistake that the idea helps you avoid.

Look for: Evaluate across varied cases

Reference example, not the only acceptable wording: Invalid-**input** handling fails both tested cases even though the overall pass fraction is 80 percent.

Project Problem and Success Test

Without opening the old lesson, explain one key idea from Project Problem and Success Test. Give an example and a mistake that the idea helps you avoid.

Look for: Define a small application purpose

Reference example, not the only acceptable wording: Suggest a suitable eligible activity for a teacher-supervised session; the teacher reviews and decides.

Evidence and Sources

Without opening the old lesson, explain one key idea from **Evidence** and Sources. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of claim support and citation evaluation

Reference example, not the only acceptable wording: Teacher explanation: Citation evaluation checks source existence, exact passage support and coverage of answer claims. A real citation can still be irrelevant.

For this classroom rubric count a claim as supported only when its cited passage entails it. Separately count correct citation links. Use the number of factual claims as the support denominator.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

After a scoring change, four old failures pass but two previous passes fail. On the original ten cases there were six passes. Compute the new total and explain what else should be reported.

Concept to explain

A **controlled improvement** changes one design choice and reruns the same tests.

Term	Meaning
controlled improvement	A revision whose effect is checked under comparable conditions rather than assumed from a changed result.
before/after	Results recorded prior to and following a change; other changes can affect the comparison.
tradeoff	A situation where improving one desired property can make another harder to achieve.
regression	In testing, a new failure introduced by a change; in modelling, prediction of a numeric target. Use the meaning indicated by the lesson.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach Week 35 as a two-session investigation using fictional data. A fictional chooser wrongly excludes an activity at the exact time limit because it uses minutes < limit.

The requirement says at most, so equality must qualify. Repair that condition, then test below, at and above the boundary plus earlier tie and invalid-**input** behavior.

Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

The change may introduce a **regression** elsewhere.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

Apply the connected idea: Testing functions and interfaces

Teach Week 33 as a two-session investigation using fictional data. Test the chooser at two scales. A **unit test** checks one validator or score calculation. An **integration test** runs validation through selection and reporting.

Write expected outcomes from the specification, not by calling the same function twice. Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

The later pipeline might still pass rejected rows to selection or mishandle its result.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

End goal for the pictorial case

By the end, I can make a focused repair and challenge it with **regression** and order-invariance tests.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Improve and Challenge using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Before: 6/10 test profiles pass.

2. Error cluster: difficult books rank too high for emerging readers.

3. Change only difficulty weight from 1 to 3.

4. After: 8/10 pass; one adventure-focused profile ranks worse—record the **tradeoff**.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

The teaching case

A repair raises success from 7/10 to 9/10 on the same validation set.

What belongs in the improvement claim?

Exact explanation to model

> **Why the result follows**
>
> Confirm fixed criteria, inspect remaining and introduced failures, then test the chosen version on fresh final cases.

Explain the added worked example

A fictional chooser wrongly excludes an activity at the exact time limit because it uses minutes < limit. The requirement says at most, so equality must qualify.

Repair that condition, then test below, at and above the boundary plus earlier tie and invalid-**input** behavior.

The repaired predicate prints 9 True, 10 True and 11 False. It assumes validated numeric durations; it is not a replacement for the schema validator.

A focused change is easier to explain and evaluate than unrelated edits made at the same time.

Key terms

Term	Meaning
Regression suite	Earlier cases rerun after a change
Change log	A record of what changed and why
Root cause	The underlying reason for a failure
Residual risk	A limitation or possible failure remaining after a repair

Apply the connected idea: Testing functions and interfaces

Test the chooser at two scales. A **unit test** checks one validator or score calculation. An **integration test** runs validation through selection and reporting.

Write expected outcomes from the specification, not by calling the same function twice.

Apply the connected idea: Testing functions and interfaces · 1

The code prints its success message only if all assertions pass. It checks the scoring component but says nothing about duration filtering, malformed records, ties or reporting. Add focused cases for those requirements.

Use the production functions from earlier weeks rather than rewriting a second copy just for tests.

The same defect can appear on both sides; it does not independently check the specified rule. Exit: 0 supplied/accepted/rejected/eligible, no selected item and a clear no-data or no-eligible explanation, without crashing.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

> **Would changing the condition to minutes <= limit + 1 correctly repair the boundary? Explain with a revealing input.**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Improve and Challenge

A fictional recommender excludes books at a page limit that is intended to include equality.

The facts we will use

Supplied facts

Contract: pages<=100; highest interest; ties choose smaller ID. Old code uses pages<100.

Supplied facts

Rows A pages100 interest5; B pages90 interest3; C pages101 interest5.

AI ACADEMY / AI-2/35

Improve and Challenge

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Identify cause

A wrongly excluded at equality, so old code selects B.

2. Focused repair

Replace <100 with <=100 without changing ranking or type checks.

3. Expected revised result

A becomes eligible and wins; C remains excluded.

4. Challenge repair

Rerun 99,100,101; invalid pages; tied scores; reversed row order.

CHANGE ONE THING

An alternative patch simply increases the limit to 101.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Identify cause	A wrongly excluded at equality, so old code selects B.
2. Focused repair	Replace <100 with <=100 without changing ranking or type checks.
3. Expected revised result	A becomes eligible and wins; C remains excluded.
4. Challenge repair	Rerun 99,100,101; invalid pages; tied scores; reversed row order.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A focused change makes the claim about what caused the error testable: the equality comparison rule caused this specific failure. **Regression** tests check surrounding behavior that should remain unchanged.

A tie case and reversed **input** order are necessary because a correct boundary does not ensure deterministic selection. Log the old failure and revised replay separately.

What this example does not establish

Keep the stated limit

Passing a small **regression suite** does not prove all behavior unchanged; record the suite's actual coverage.

Change one thing in the pictured case

Change one thing

Change: An alternative patch simply increases the limit to 101.

Predicted result

Prediction: C is wrongly admitted as well as A.

Why the result follows

Reason: Changing the maximum alters the contract instead of repairing its comparison.

Ask while showing the picture

Ask while showing the picture · 2

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Improve and Challenge, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: A repair raises success from 7/10 to 9/10 on the same validation set.

2. State the decision to explain

Write the question to explain: What belongs in the improvement claim?

3. Reveal the worked reasoning

Reveal after **prediction**s: Confirm fixed criteria, inspect remaining and introduced failures, then test the chosen version on fresh final cases.

Hold back the independent task

Keep for an independent attempt: After changing a scoring weight, the system fixes one request but breaks two previously correct ones. What should the improvement record say?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

No. It would incorrectly accept 11 when the limit is 10. Exit: The strict comparison excluded the boundary; changed < to <=; below/at/above and prior checks rerun; finite tests and untested situations remain.

Accept equivalent specific records.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.

Stage	Teacher action
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Teach the connected idea

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Completion practice

Completion task: For limit 15, give expected eligibility at 14, 15 and 16.

Reasoning to listen for

Teacher check: True, True, False.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Improving without hiding failures. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: A clean **regression** run does not erase the earlier failure. Keep **before/after evidence** and remaining limitations.

If a fix was informed by reserved evaluation cases, report that contamination and obtain new independent cases.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Connected teaching cycle: Testing functions and interfaces

Target: Test functions and their connections

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: What expected score should a test use for fun 4 and learning 3?

Reasoning to listen for

Teacher check: 11 from 2 * 4 + 3, calculated independently.

Individual check: For Testing functions and interfaces, use its worked case to evaluate this proposed explanation: “A successful run proves the conclusion”. Explain the deciding step.

Expected repair

Expected correction: Test results establish only the executed cases on the tested version. Retain failures and version information.

Use explicit expected rejection behavior for invalid data instead of assuming every exception is acceptable.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

Pictorial practice question

Why are tests at both 100 and 101 needed?

Reasoned teacher answer

100 confirms equality is accepted; 101 confirms values above the unchanged maximum remain rejected.

Changed-case reasoning: C is wrongly admitted as well as A. Changing the maximum alters the contract instead of repairing its comparison.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Improve and Challenge investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Choose one **evidence**-backed change, freeze the test set, predict effects, rerun every case, compare 6/10 to 8/10, and inspect **regression**s.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome

Record	What to capture
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A successful run proves the conclusion: A clean **regression** run does not erase the earlier failure. Keep **before/after evidence** and remaining limitations.

If a fix was informed by reserved evaluation cases, report that contamination and obtain new independent cases.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: rating validity should include 0 through 5 but a bug uses 0 < rating < 5.

Repair to inclusive bounds; test -1,0,1,5,6 giving False,True,True,True,False.

A convenient shortcut preserves meaning: Changing the expected boundary result to False would make the test pass while violating the requirement.

Keep the requirement and independently calculated expectation; repair the implementation. If requirements truly change, document that as a policy change.

Check these misconceptions

A successful run proves the conclusion: Test results establish only the executed cases on the tested version. Retain failures and version information.

Use explicit expected rejection behavior for invalid data instead of assuming every exception is acceptable.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: X valid (minutes 8, fun 3, learning 3), Y valid (9,4,1), Z invalid missing fun, limit 9.

X/Y tie at 9; X wins; counts 3/2/1/2.

A convenient shortcut preserves meaning: A test suite that only checks the winner name can miss wrong rejection counts. Check the full expected report: supplied, accepted, rejected, eligible, selected and reason.

Avoid relying on dictionary display order when the contract concerns values.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Improve and Challenge questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Confirm fixed criteria, inspect remaining and introduced failures, then test the chosen version on fresh final cases.

Independent question

Independent question · withhold the key

After changing a scoring weight, the system fixes one request but breaks two previously correct ones. What should the improvement record say?

Expected independent answer

Teacher key · Expected independent answer

Report the fixed and newly broken cases, compare the same test set, and decide using the stated success criteria. One repaired example alone does not prove overall improvement.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Fixing one example proves a change is beneficial overall.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

> **Teacher key · Original transfer answer**
>
> New total is 6+4−2=8/10. Report which four failures were fixed and which two **regression**s appeared, along with the exact change.
>
> Use a fresh final set after selecting the version; repeatedly tuning on ten cases makes them validation data.

Explanations for the additional workbook examples

W1 Locate the cause

> **Workbook question · W1**
>
> Why does minutes < limit reject a valid exactly-at-limit activity?

> **Teacher reasoning · W1**
>
> Strict less-than excludes equality; at most requires <=.

Teacher reasoning · W1

If minutes equals limit, minutes < limit evaluates to False because the comparison is strict. The phrase at most includes equality, so the validated-duration predicate must be minutes <= limit.

Test below, at and above the limit to reveal this difference.

W2 Test neighbors

Workbook question · W2

For limit 15, give expected eligibility at 14, 15 and 16.

Teacher reasoning · W2

True, True, False.

Teacher reasoning · W2

With limit 15, compare each value using <=: 14 <= 15 is True, 15 <= 15 is True, and 16 <= 15 is False. Keep equality visible as the boundary case rather than testing only clearly short and clearly long activities.

W3 Reject a bad fix

Workbook question · W3

A proposed fix accepts minutes <= limit + 1. Give a counterexample at limit 15.

Teacher reasoning · W3

It accepts 16, which should be excluded.

Teacher reasoning · W3

Set minutes to 16 and limit to 15. The proposed condition checks 16 <= 16 and accepts it, but the requirement checks 16 <= 15 and rejects it.

This single counterexample demonstrates that adding one to the limit changed the policy rather than repairing it.

W4 Repair independently

Workbook question · W4

Write the eligibility predicate for at most **12 minutes** and a **regression** set including 11,12,13 and 0. Add one chooser tie case and one invalid-data case to rerun.

Teacher reasoning · W4

Use minutes <= limit after validation. Expectations True, True, False, True. Include a deterministic tie case and a schema-invalid row that never reaches selection; state expected outcomes.

W5 Write the change log

Workbook question · W5

Record the bug, change, **evidence** and one remaining limit without claiming all bugs are gone.

Teacher reasoning · W5

The strict comparison excluded the boundary; changed < to <=; below/at/above and prior checks rerun; finite tests and untested situations remain. Accept equivalent specific records.

Testing functions and interfaces — W1 Choose test scale

Classify testing score(3,2) versus running raw rows through the whole chooser.

The first is a unit check of scoring; the second is an integration check.

Testing functions and interfaces — W2 Calculate an oracle

What expected score should a test use for fun 4 and learning 3?

11 from 2 * 4 + 3, calculated independently.

Calculate the expected score from the **written rule**: 2*4 + 3 = 11. Record 11 before running score.

Computing the expectation by calling the same score function would allow a shared implementation error to appear on both sides of the comparison.

Testing functions and interfaces — W3 Repair a weak test

A test compares recommend(items,10) to itself. Replace it with a useful assertion.

Use a controlled **fixture** and an independently specified expected winner, plus relevant report counts and reason.

Testing functions and interfaces — W4 Test independently

Design an integration **fixture**: A valid (minutes 10, fun 2, learning 4), B valid (10,3,2), C invalid (5,9,1), limit 10. State winner, counts and a reversed-order expectation.

3 supplied, 2 accepted, 1 rejected, 2 eligible. A and B score 8; A wins alphabetically. Reversing order preserves A and the counts.

Testing functions and interfaces — W5 Cover absence

> What expected result should an empty-**input** integration case specify?

0 supplied/accepted/rejected/eligible, no selected item and a clear no-data or no-eligible explanation, without crashing.

Assess independent transfer

> **Independent question · withhold the key**
>
> Selected case: Write the eligibility predicate for at most **12 minutes** and a **regression** set including 11,12,13 and 0. Add one chooser tie case and one invalid-data case to rerun.

Selected case answer

> **Teacher key · Selected case answer**
>
> Use minutes <= limit after validation. Expectations True, True, False, True. Include a deterministic tie case and a schema-invalid row that never reaches selection; state expected outcomes.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Improve and Challenge to Transparent Book Recommender, adapt support and plan a later changed-case check.

Transparent Book Recommender

Publish the cause, one change, fixed test set, **before/after evidence**, and remaining limitation.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 36: Capstone Expo. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Compare a requirement sentence and one condition side by side. Extension: introduce one deliberate mutation and explain which **regression** case detects it.

Independent W4 **evidence**: Use minutes <= limit after validation. Expectations True, True, False, True. Include a deterministic tie case and a schema-invalid row that never reaches selection; state expected outcomes.

Record whether the learner needed no help, a hint or explanation. Fresh retry: rating validity should include 0 through 5 but a bug uses 0 < rating < 5.

Repair to inclusive bounds; test -1,0,1,5,6 giving False,True,True,True,False.

Next week: Defending the data application. Reuse this week's explicit contracts and **evidence**.

Readiness and reassessment for Testing functions and interfaces

Provide a requirements-to-tests table. Extension: deliberately change the scoring multiplier and confirm a revealing test fails before restoring the implementation.

Independent W4 **evidence**: 3 supplied, 2 accepted, 1 rejected, 2 eligible. A and B score 8; A wins alphabetically. Reversing order preserves A and the counts. Record whether the learner needed no help, a hint or explanation.

Fresh retry: X valid (minutes 8, fun 3, learning 3), Y valid (9,4,1), Z invalid missing fun, limit 9. X/Y tie at 9; X wins; counts 3/2/1/2.

Plan the delayed return

Suggested return points in this level: ai-2/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For each recommender revision, retain cause, code change, targeted test, **regression** results and unresolved issues.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Improve and Challenge lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

> Teaching target: Repair a specific observed failure

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

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>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Before class, solve the supported practice: For limit 15, give expected eligibility at 14, 15 and 16.

Supported practice answer

Reasoned practice answer: True, True, False.

Check every learner before discussion: Use the worked case for Improving without hiding failures. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: A clean **regression** run does not erase the earlier failure. Keep **before/after evidence** and remaining limitations.

If a fix was informed by reserved evaluation cases, report that contamination and obtain new independent cases.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: An alternative patch simply increases the limit to 101. Require a **prediction** and a reason before comparing. Expected result: C is wrongly admitted as well as A.

Reason: Changing the maximum alters the contract instead of repairing its comparison.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Improve and Challenge graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Improve and Challenge: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional recommender excludes books at a page limit that is intended to include equality. Supplied facts: Contract: pages<=100; highest interest; ties choose smaller ID. Old code uses pages<100.

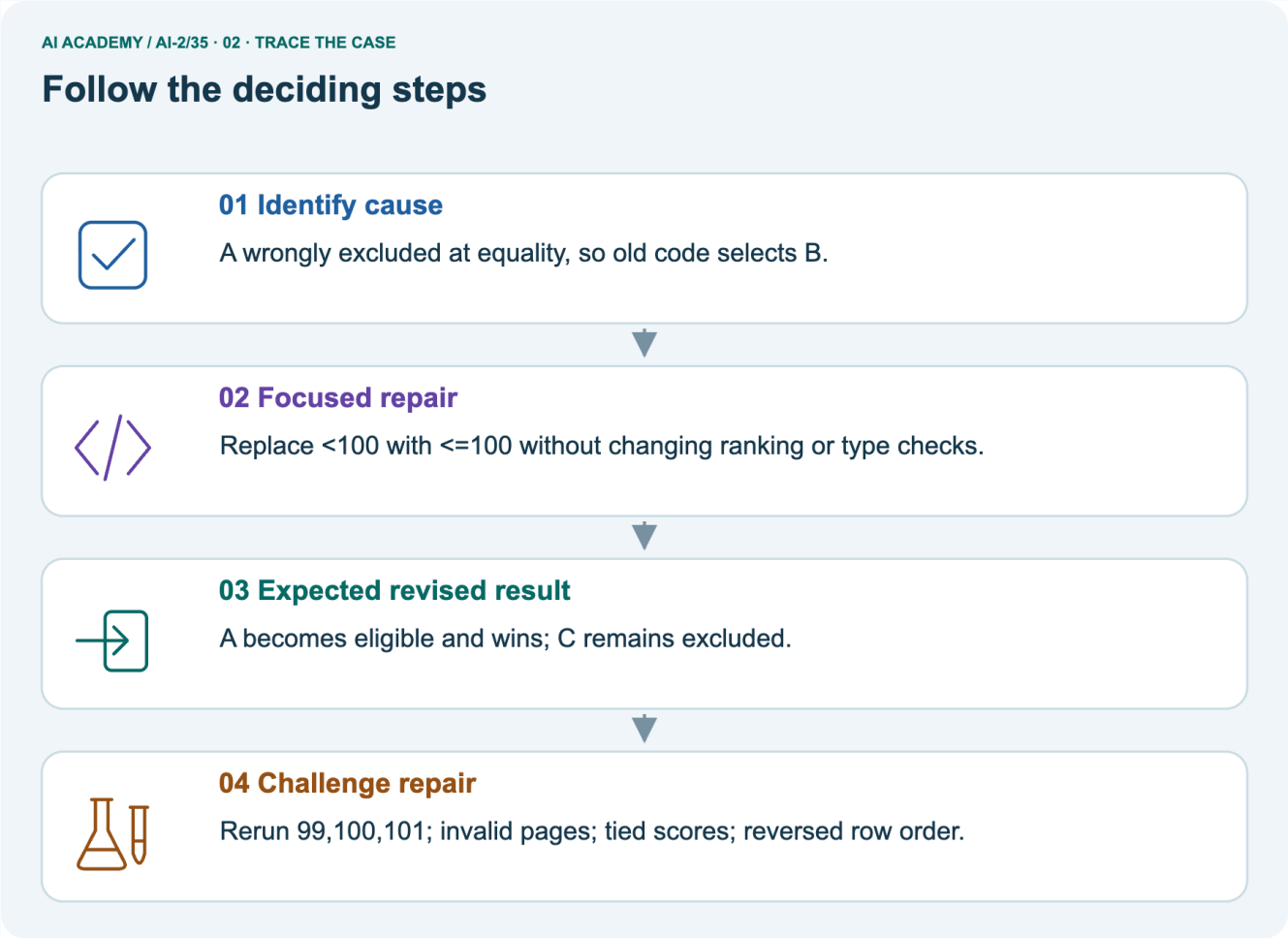
Rows A pages100 interest5; B pages90 interest3; C pages101 interest5.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

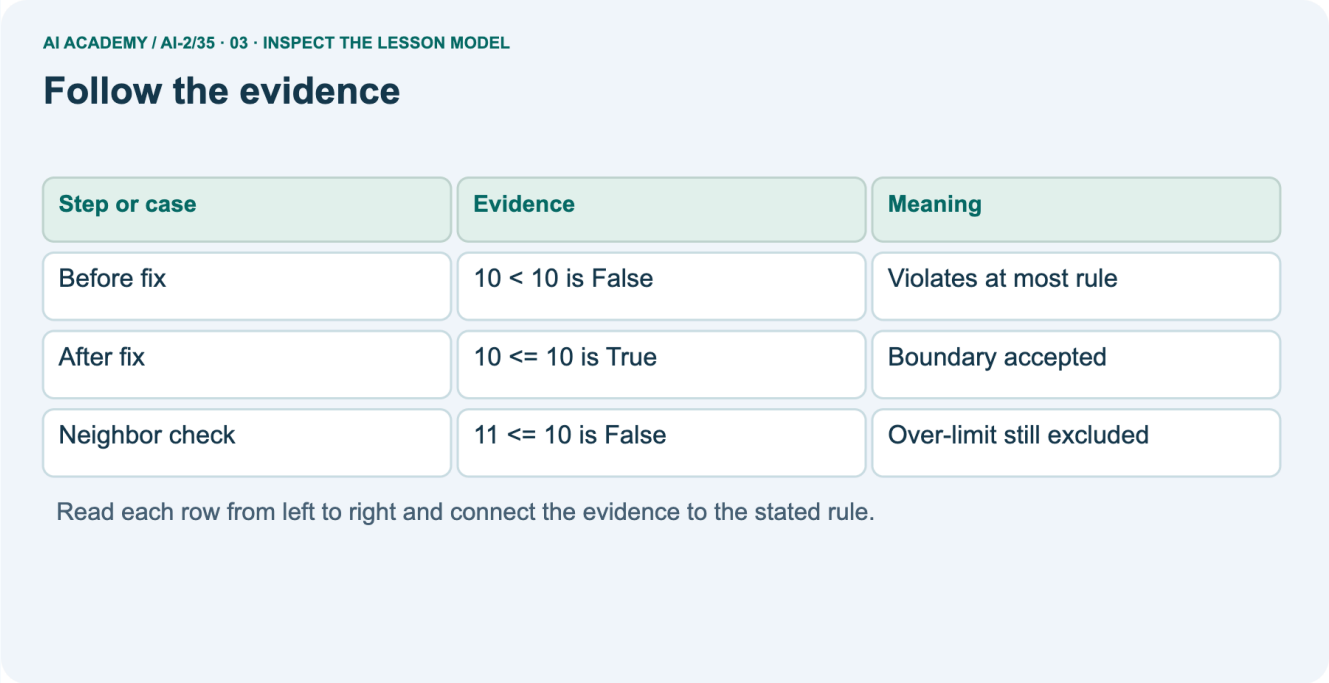
Identify cause: A wrongly excluded at equality, so old code selects B. → Focused repair: Replace <100 with <=100 without changing ranking or type checks. → Expected revised result: A becomes eligible and wins; C remains excluded. → Challenge repair: Rerun 99,100,101; invalid pages; tied scores; reversed row order.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Follow the evidence



Follow the labelled picture with the matching explanation. Open the vector diagram

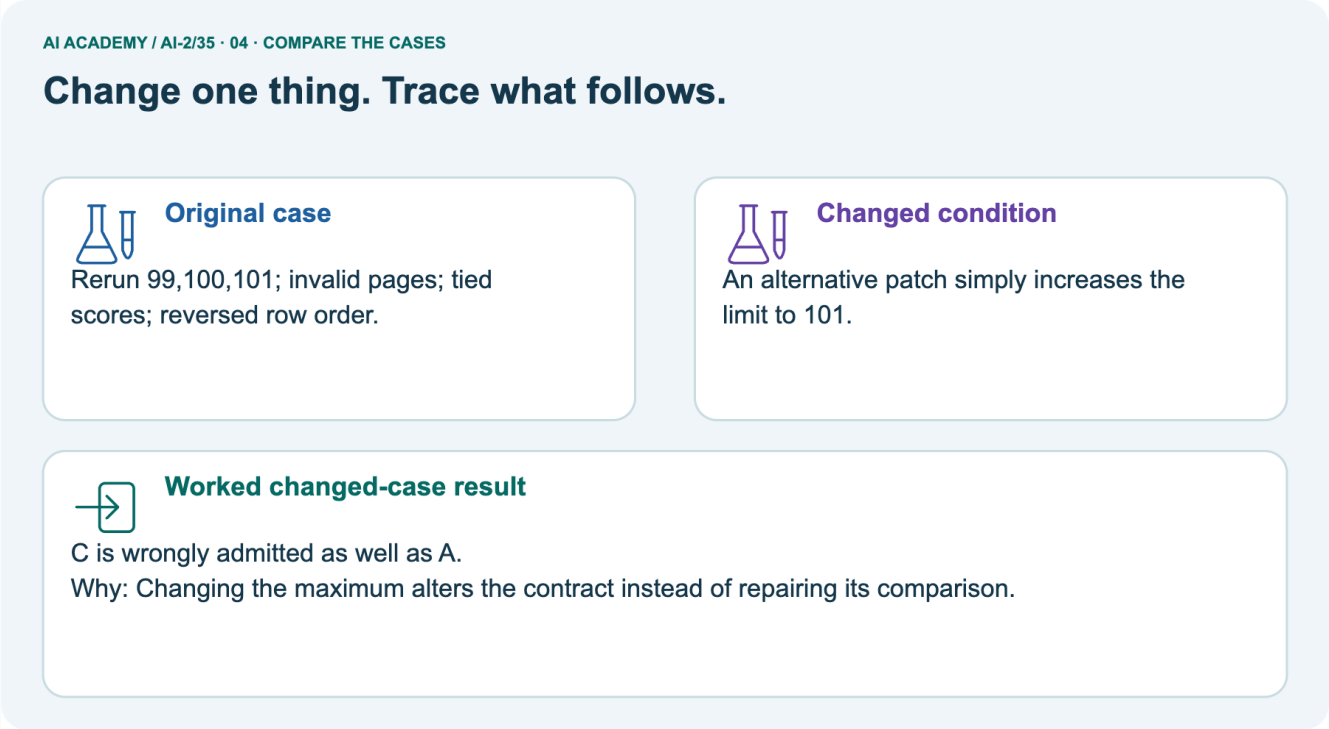
Follow the **evidence**. Before fix | 10 < 10 is False | Violates at most rule. After fix | 10 <= 10 is True | Boundary accepted. Neighbor check | 11 <= 10 is False | Over-limit still excluded

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Rerun 99,100,101; invalid pages; tied scores; reversed row order. Changed condition: An alternative patch simply increases the limit to 101. Expected result: C is wrongly admitted as well as A.

Reason: Changing the maximum alters the contract instead of repairing its comparison.

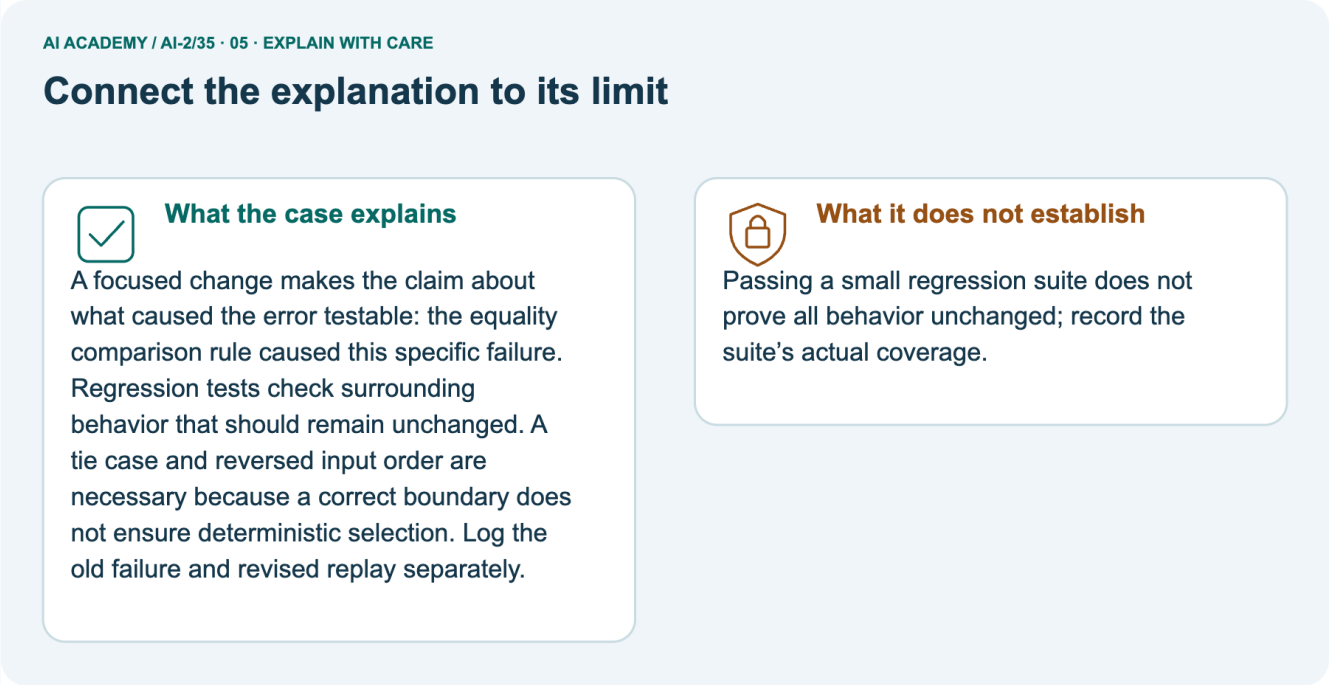
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Changing the maximum alters the contract instead of repairing its comparison.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A focused change makes the claim about what caused the error testable: the equality comparison rule caused this specific failure. **Regression** tests check surrounding behavior that should remain unchanged.

A tie case and reversed **input** order are necessary because a correct boundary does not ensure deterministic selection. Log the old failure and revised replay separately.

Limit: Passing a small **regression suite** does not prove all behavior unchanged; record the suite's actual coverage.

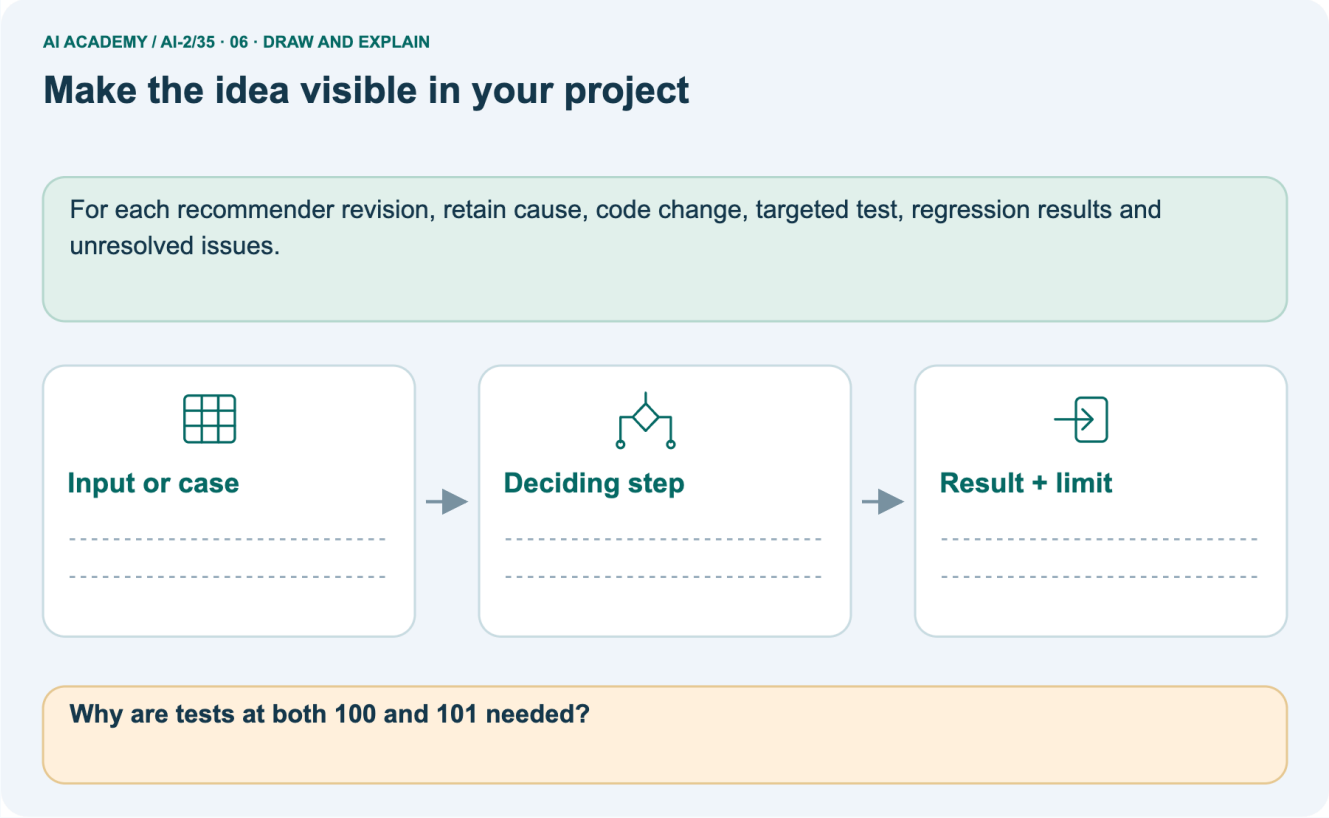
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Passing a small **regression suite** does not prove all behavior unchanged; record the suite's actual coverage.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For each recommender revision, retain cause, code change, targeted test, **regression** results and unresolved issues. Draw your **input**, deciding step, result and limit. Supported question: Why are tests at both 100 and 101 needed?

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: 100 confirms equality is accepted; 101 confirms values above the unchanged maximum remain rejected.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.